

MUHAMMAD HAMMAD ASHER

AI Engineer · Machine Learning Engineer · Software Engineer

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| hammadasher100.github.io/Portfolio

PROFESSIONAL SUMMARY

Techathon 2.0 national winner (Agentic AI Track, 2026) and Software Engineering student at Bahria University Karachi with hands-on experience building and deploying AI systems that ship to production. Experienced with agentic AI, MLOps pipelines, and full-stack deployment. Passionate about building practical AI solutions that solve real problems and deliver lasting value.

EDUCATION

Bahria University, Karachi - B.S. Software Engineering | CGPA: **3.63 / 4.00** *Expected May 2028*

Relevant: Data Structures & Algorithms, Operating Systems, DBMS, Software Design & Architecture, OOP, DAA

Saylani Mass IT Training (SMIT) - AI & Data Science

July 2025 – Present

AWARDS & HONORS

1st Place - Agentic AI Track, Techathon 2.0 github.com/hammadAsher100/Agentic_ai_Physics_law *2026*

- Designed a 4-stage multi-agent GPT-4o pipeline that detects fabricated sensor readings via adversarial physics-law cross-examination; won 1st place competing against university teams nationwide.

Runner-Up - The Heist, Procom 2025 (FAST NUCES) *2025*

- Placed 2nd in a nationally competitive algorithmic problem-solving event at one of Pakistan's top engineering universities.

PROJECTS

Bank Marketing Term Deposit Prediction - MLOps Pipeline [GitHub](#) [Live App](#)

Python · scikit-learn · MLflow · DVC · FastAPI · PySpark · Docker · Streamlit · GitHub Actions

- Engineered a production-grade MLOps pipeline on 45,211-row UCI dataset, DVC orchestration, Great Expectations validation, SMOTE balancing, and MLflow tracking, achieving 92% F1 score.
- Containerised FastAPI + Streamlit with Docker; CI/CD gate blocks deployment if $F1 < 0.60$; Evidently AI drift detection ($PSI > 0.2$) triggers automated weekly retraining with zero-downtime inference.

MediCore HMS - Hospital Management System [GitHub](#) [Live Demo](#)

Python · Flask · PostgreSQL · Flask-Login · bcrypt · JWT · GitHub Actions · Render

- Architected and deployed a full-stack RBAC web app (4 roles, 6+ modules) with real-time appointment conflict detection, pharmacy inventory alerts, invoice generation, and a live admin analytics dashboard on Render.
- Secured with bcrypt + Flask-Login auth; GitHub Actions CI pipeline runs automated tests on every push and gates deployments to Render.

- Weather & Finance ETL Analytics Dashboard** [GitHub](#) [Live App](#)

Python · Streamlit · pandas · Plotly · OpenWeatherMap API · Alpha Vantage API

- Built a real-time ETL pipeline across 15,000+ global entities; Plotly dashboards surface financial indicators (MAs, Volume) and atmospheric metrics, deployed on Streamlit Cloud with zero-downtime access.

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, JavaScript, SQL

ML & MLOps: scikit-learn, TensorFlow, MLflow, DVC, Evidently AI, Great Expectations, PySpark, CNNs, RNNs

LLM & Agentic AI: OpenAI GPT-4o, Multi-Agent Architecture, Prompt Engineering, Prompt Injection Defence, LangChain

Data Visualisation: Plotly, Streamlit, Matplotlib, pandas, NumPy, ETL Pipelines

Backend & APIs: Flask, FastAPI, RESTful API Design, JWT, bcrypt, Flask-Login

Deployment & DevOps: Docker, Kubernetes, GitHub Actions, Render, Streamlit Cloud, AWS (S3, EC2)

Databases: PostgreSQL, MySQL, SQL Server

Concepts: OOP, DSA, Design Patterns (MVC, Factory, Observer, Singleton, Decorator), SDLC, Agile